



Name : Azaria Cindy Sahasika
Number Id : 2341760169 / 06
Class : 2G – Business Information System
Lesson : Advanced Web Programing
Github Link : <https://github.com/azariacindy>

Jobsheet 3 Database Analytical

Tujuan Praktikum:

Setelah melakukan praktikum ini, mahasiswa diharapkan dapat lebih mengenal table dimensi, table fakta dan apa itu OLAP.

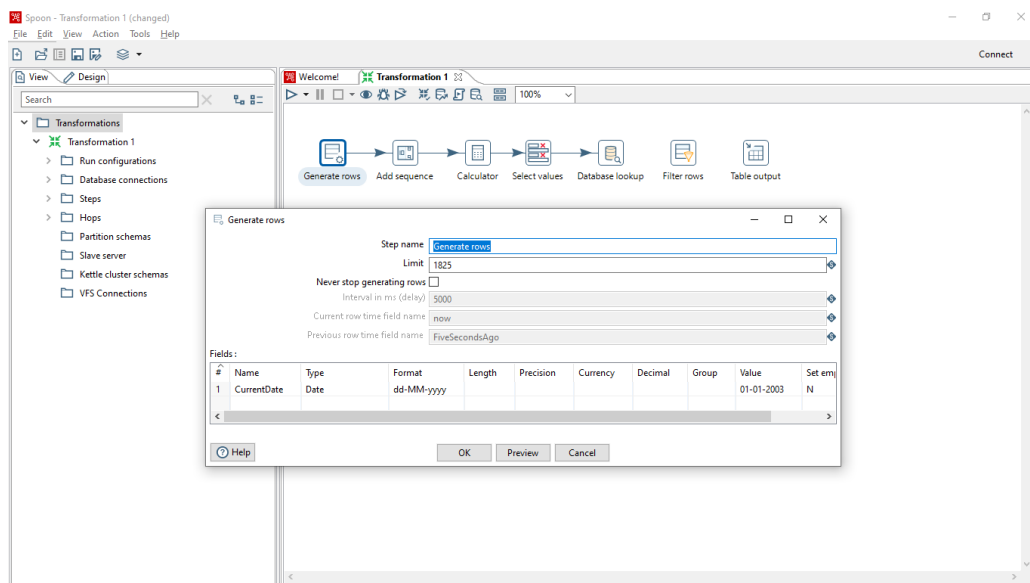
Studi Kasus:

Berdasarkan studi kasus sebelumnya, Pimpinan dari LegendVehicle mulai mengenal teknologi dan berpandangan visioner. LegendVehicle akan menerapkan data warehouse pada proses bisnisnya untuk menganalisa proses transaksi yang dilihat dari pembayaran yang masuk. Proses ini digunakan untuk menganalisa KPI "jumlah omset yang dimiliki".

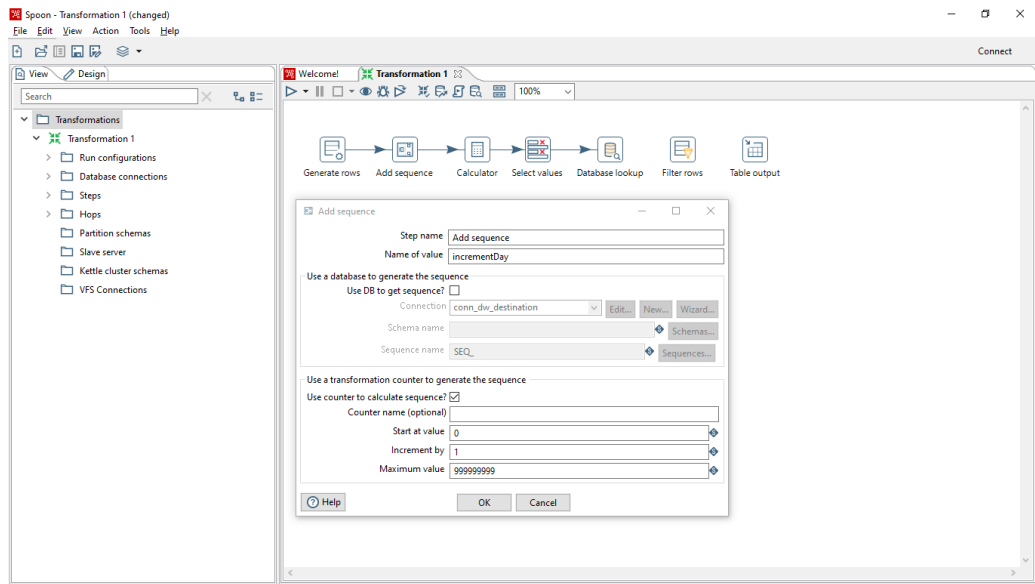
Untuk membentuk data warehouse tersebut, Data Engineer pada LegendVehicle perlu membuat sebuah database yang digunakan sebagai databasae OLAP. Data dari OLTP yang ada akan di "ETL" kan menuju database OLAP.

A. Dimensi Waktu

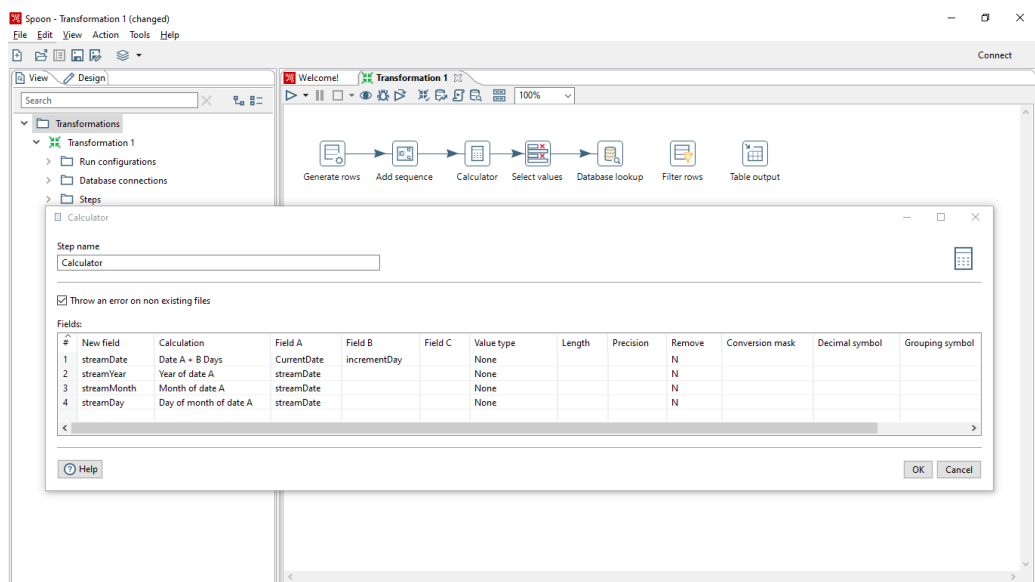
1. Konfigurasi generate rows (membuat data baru)



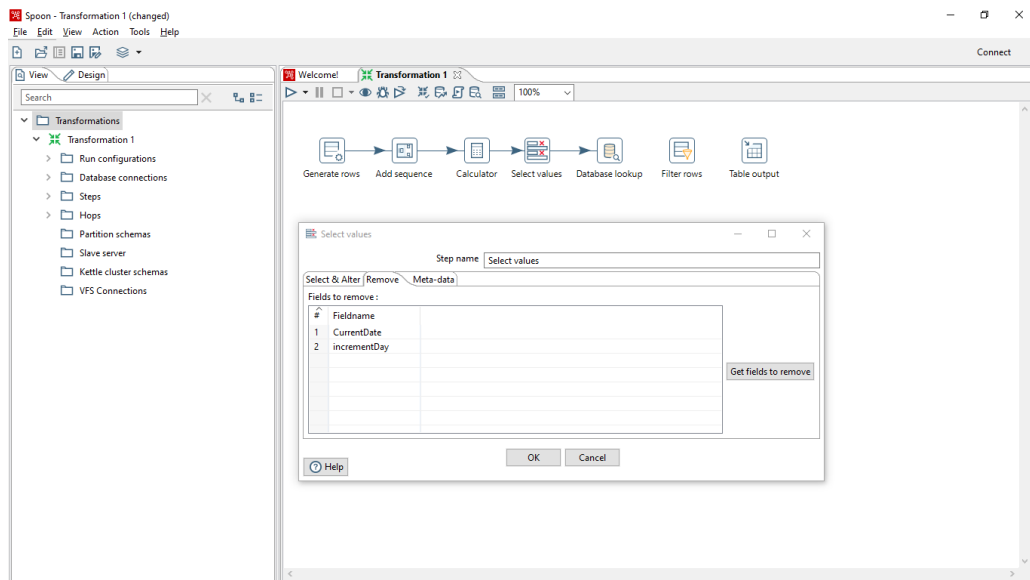
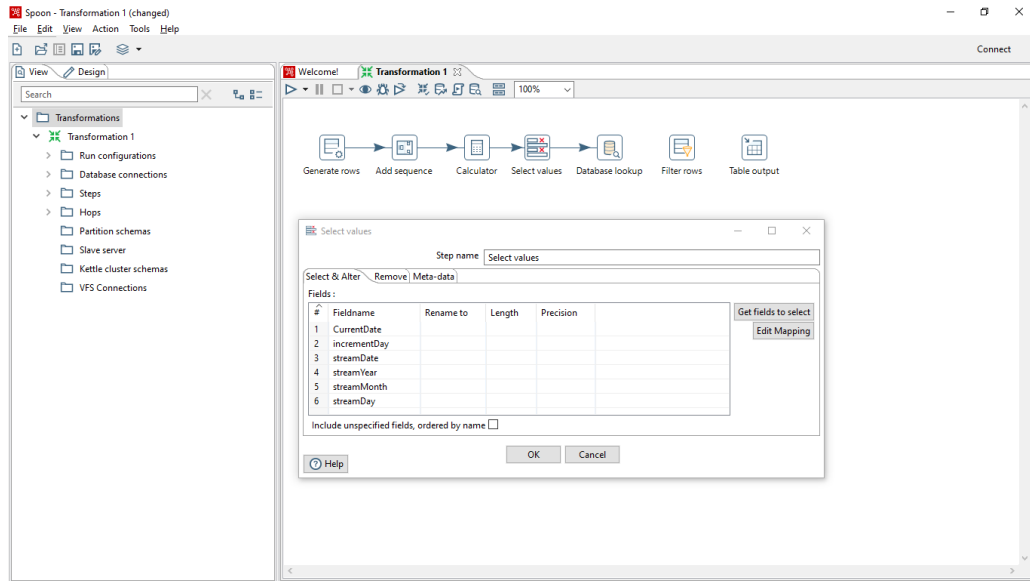
2. Konfigurasi add sequence (menambahkan ID urut)



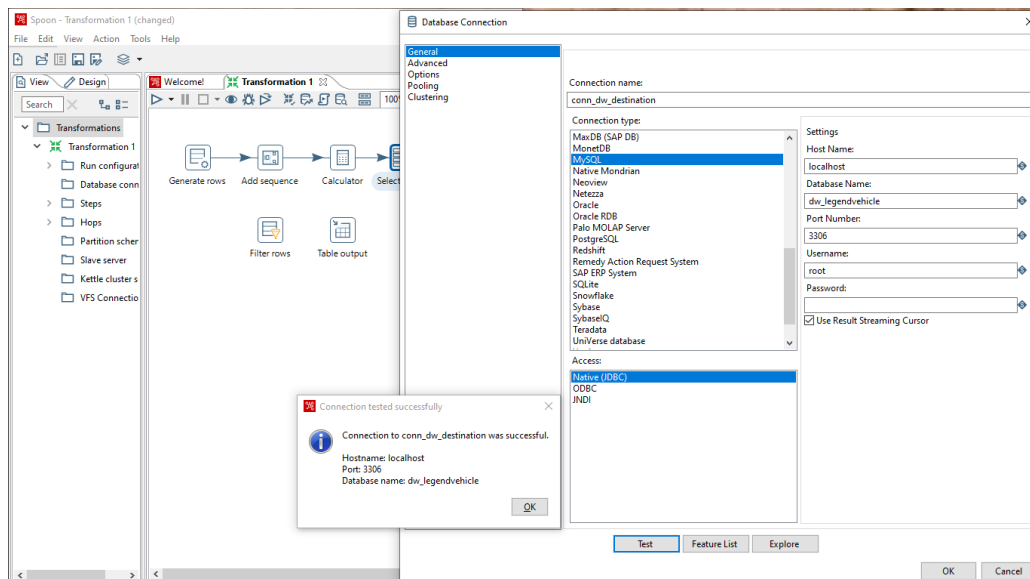
3. Konfigurasi calculator (melakukan perhitungan)



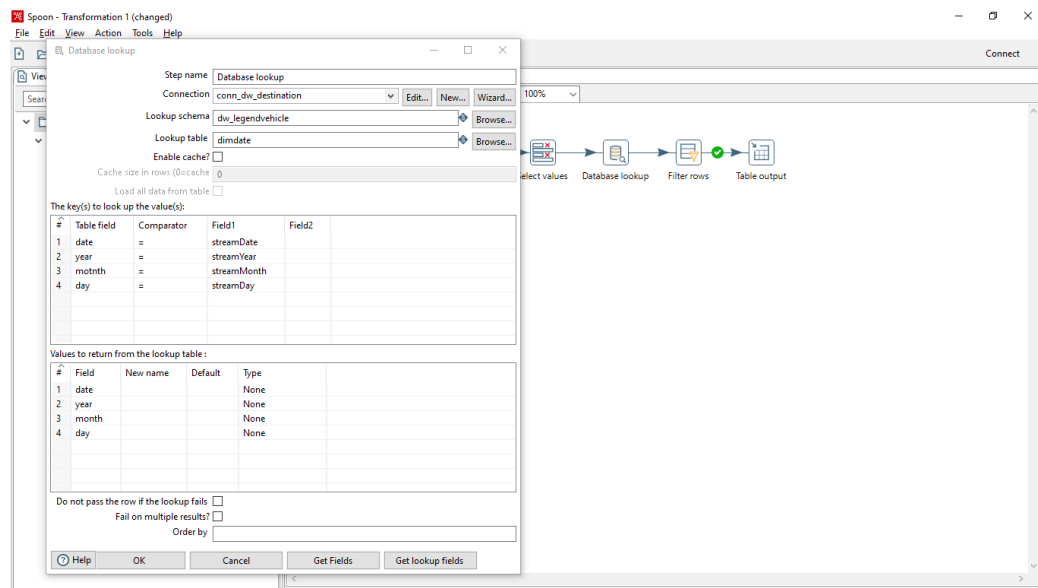
4. Konfigurasi select values (menyaring kolom yang tidak dibutuhkan)



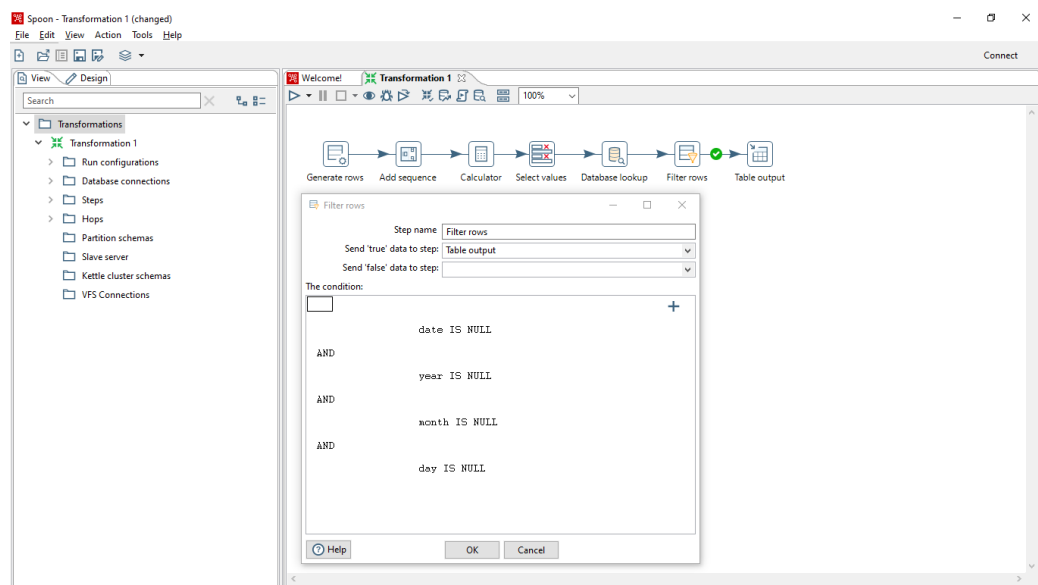
5. Database connection (menhubungkan koneksi dari pentaho ke mysql)



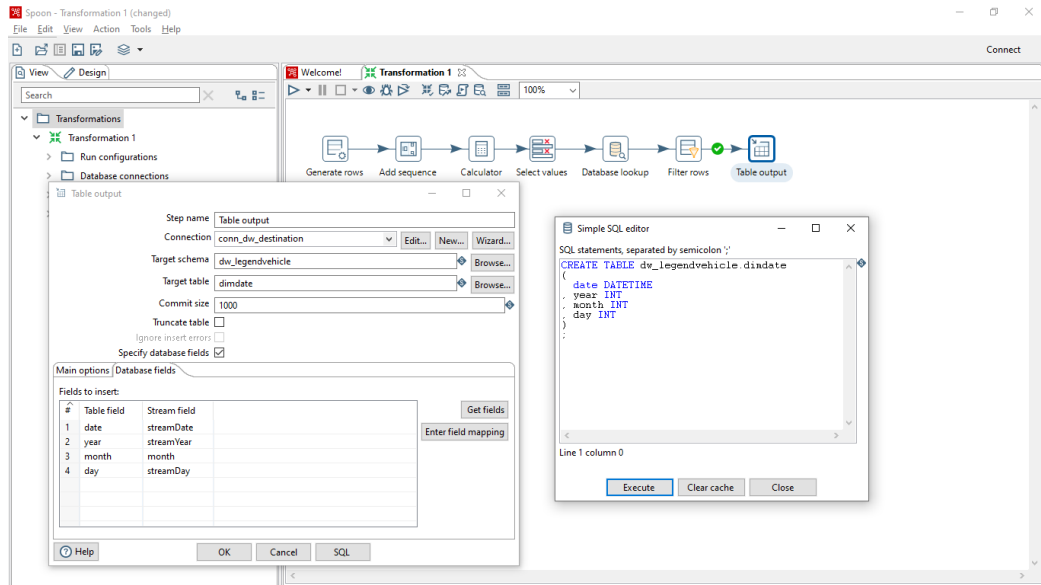
6. Konfigurasi database lookup (melakukan lookup ke database)



7. Konfigurasi filter rows (menyaring baris sesuai dengan kondisi)



8. Konfigurasi table output (menyimpan hasil akhir ke dalam database)



➔ Check the contents of the dimdate table

Server: localhost » Database: dw_legendvehicle » Table: dimdate

Table structure

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	date	datetime			Yes	NULL			Change Drop More
2	year	int			Yes	NULL			Change Drop More
3	month	int			Yes	NULL			Change Drop More
4	day	int			Yes	NULL			Change Drop More

Check all With selected: Browse Change Drop Primary Unique Index Spatial Fulltext

date	year	month	day
2003-01-13 00:00:00	2003	1	13
2003-01-14 00:00:00	2003	1	14
2003-01-15 00:00:00	2003	1	15
2003-01-16 00:00:00	2003	1	16
2003-01-17 00:00:00	2003	1	17
2003-01-18 00:00:00	2003	1	18
2003-01-19 00:00:00	2003	1	19
2003-01-20 00:00:00	2003	1	20
2003-01-21 00:00:00	2003	1	21
2003-01-22 00:00:00	2003	1	22
2003-01-23 00:00:00	2003	1	23
2003-01-24 00:00:00	2003	1	24
2003-01-25 00:00:00	2003	1	25

Tugas 1:

Buka preview tab pada execution result area di setiap proses object. amati input dan output data yang ada. bandingkan di setiap prosesnya. jelaskan perbedaan di setiap prosesnya

Object Process	Input	Output
Generate rows	Tidak ada input karena generate rows digunakan untuk membuat data baru.	Data yang dihasilkan berupa baris baru dengan nilai tetap atau acak berdasarkan konfigurasi.
Add sequence	Data dari hasil generate rows sebelumnya.	Data dengan tambahan kolom urutan.
Calculator	Data yang masuk dari add sequence sebelumnya atau data dengan nilai mentah.	Data dengan hasil perhitungan konfigurasi.
Select values	Data atau semua kolom yang masuk dari proses sebelumnya.	Kolom yang diseleksi atau dipilih sesuai kebutuhan saja.
Database lookup	Data hasil dari select values, proses ini memanfaatkan nilai dari kolom tertentu untuk mencari data di database.	Mendapatkan nilai tambahan dari database ke dalam dataset berdasarkan hasil lookup.
Filter rows	Data hasil dari proses database lookup atau semua baris.	Hanya menampilkan baris data yang memenuhi syarat atau kondisi filter.
Table output	Data dari proses filter rows.	Data disimpan dalam table pada database.

1. Generate rows

The screenshot shows the SAP Spoon - Transformation 1 interface. The workflow consists of seven steps: Generate rows, Add sequence, Calculator, Select values, Database lookup, Filter rows, and Table output. The 'Generate rows' step is highlighted with a green checkmark. The 'Execution Results' pane at the bottom shows a table with 21 rows, all containing the date '01-01-2003' in the 'CurrentDate' column.

#	CurrentDate
1	01-01-2003
2	01-01-2003
3	01-01-2003
4	01-01-2003
5	01-01-2003
6	01-01-2003
7	01-01-2003
8	01-01-2003
9	01-01-2003
10	01-01-2003
11	01-01-2003
12	01-01-2003
13	01-01-2003
14	01-01-2003
15	01-01-2003
16	01-01-2003
17	01-01-2003
18	01-01-2003
19	01-01-2003
20	01-01-2003
21	01-01-2003

2. Add sequence

The screenshot shows the SAP Spoon - Transformation 1 interface. The workflow consists of seven steps: Generate rows, Add sequence, Calculator, Select values, Database lookup, Filter rows, and Table output. The 'Add sequence' step is highlighted with a green checkmark. The 'Execution Results' pane at the bottom shows a table with 21 rows, each containing a date in the 'CurrentDate' column and an increment value in the 'incrementDay' column.

#	CurrentDate	incrementDay
1	01-01-2003	0
2	01-01-2003	1
3	01-01-2003	2
4	01-01-2003	3
5	01-01-2003	4
6	01-01-2003	5
7	01-01-2003	6
8	01-01-2003	7
9	01-01-2003	8
10	01-01-2003	9
11	01-01-2003	10
12	01-01-2003	11
13	01-01-2003	12
14	01-01-2003	13
15	01-01-2003	14
16	01-01-2003	15
17	01-01-2003	16
18	01-01-2003	17
19	01-01-2003	18
20	01-01-2003	19
21	01-01-2003	20

3. Calculator

The screenshot shows the SAP Spoon - Transformation 1 interface. The main canvas displays a sequence of steps: Generate rows, Add sequence, Calculator, Select values, Database lookup, Filter rows, and Table output. The 'Execution Results' window is open, showing the 'Step Metrics' tab. The 'Calculator' step is selected, and the 'Preview data' tab is active. The data table shows the following columns: #, CurrentDate, incrementDay, streamDate, streamYear, streamMonth, and streamDay. The data is as follows:

#	CurrentDate	incrementDay	streamDate	streamYear	streamMonth	streamDay
1	01-01-2003	0	2003/01/01 00:00:00.000	2003	1	1
2	01-01-2003	1	2003/01/02 00:00:00.000	2003	1	2
3	01-01-2003	2	2003/01/03 00:00:00.000	2003	1	3
4	01-01-2003	3	2003/01/04 00:00:00.000	2003	1	4
5	01-01-2003	4	2003/01/05 00:00:00.000	2003	1	5
6	01-01-2003	5	2003/01/06 00:00:00.000	2003	1	6
7	01-01-2003	6	2003/01/07 00:00:00.000	2003	1	7
8	01-01-2003	7	2003/01/08 00:00:00.000	2003	1	8
9	01-01-2003	8	2003/01/09 00:00:00.000	2003	1	9
10	01-01-2003	9	2003/01/10 00:00:00.000	2003	1	10
11	01-01-2003	10	2003/01/11 00:00:00.000	2003	1	11
12	01-01-2003	11	2003/01/12 00:00:00.000	2003	1	12
13	01-01-2003	12	2003/01/13 00:00:00.000	2003	1	13
14	01-01-2003	13	2003/01/14 00:00:00.000	2003	1	14
15	01-01-2003	14	2003/01/15 00:00:00.000	2003	1	15
16	01-01-2003	15	2003/01/16 00:00:00.000	2003	1	16
17	01-01-2003	16	2003/01/17 00:00:00.000	2003	1	17
18	01-01-2003	17	2003/01/18 00:00:00.000	2003	1	18
19	01-01-2003	18	2003/01/19 00:00:00.000	2003	1	19
20	01-01-2003	19	2003/01/20 00:00:00.000	2003	1	20
21	01-01-2003	20	2003/01/21 00:00:00.000	2003	1	21

4. Select values

The screenshot shows the SAP Spoon - Transformation 1 interface. The main canvas displays a sequence of steps: Generate rows, Add sequence, Calculator, Select values, Database lookup, Filter rows, and Table output. The 'Execution Results' window is open, showing the 'Step Metrics' tab. The 'Select values' step is selected, and the 'Preview data' tab is active. The data table shows the following columns: #, streamDate, streamYear, streamMonth, and streamDay. The data is as follows:

#	streamDate	streamYear	streamMonth	streamDay
1	2003/01/01 00:00:00.000	2003	1	1
2	2003/01/02 00:00:00.000	2003	1	2
3	2003/01/03 00:00:00.000	2003	1	3
4	2003/01/04 00:00:00.000	2003	1	4
5	2003/01/05 00:00:00.000	2003	1	5
6	2003/01/06 00:00:00.000	2003	1	6
7	2003/01/07 00:00:00.000	2003	1	7
8	2003/01/08 00:00:00.000	2003	1	8
9	2003/01/09 00:00:00.000	2003	1	9
10	2003/01/10 00:00:00.000	2003	1	10
11	2003/01/11 00:00:00.000	2003	1	11
12	2003/01/12 00:00:00.000	2003	1	12
13	2003/01/13 00:00:00.000	2003	1	13
14	2003/01/14 00:00:00.000	2003	1	14
15	2003/01/15 00:00:00.000	2003	1	15
16	2003/01/16 00:00:00.000	2003	1	16
17	2003/01/17 00:00:00.000	2003	1	17
18	2003/01/18 00:00:00.000	2003	1	18
19	2003/01/19 00:00:00.000	2003	1	19
20	2003/01/20 00:00:00.000	2003	1	20
21	2003/01/21 00:00:00.000	2003	1	21

5. Database lookup

The screenshot shows the 'Spoon - Transformation 1' window. The 'Execution Results' tab is active, displaying a table with 21 rows of data. The table has columns: #, streamDate, streamYear, streamMonth, streamDay, date, year, month, and day. The data represents a sequence of dates from 2003/01/01 to 2003/01/21. The 'Database lookup' step is highlighted in the process flow.

#	streamDate	streamYear	streamMonth	streamDay	date	year	month	day
1	2003/01/01 00:00:00.000	2003	1	1	<null>	<null>	<null>	<null>
2	2003/01/02 00:00:00.000	2003	1	2	<null>	<null>	<null>	<null>
3	2003/01/03 00:00:00.000	2003	1	3	<null>	<null>	<null>	<null>
4	2003/01/04 00:00:00.000	2003	1	4	<null>	<null>	<null>	<null>
5	2003/01/05 00:00:00.000	2003	1	5	<null>	<null>	<null>	<null>
6	2003/01/06 00:00:00.000	2003	1	6	<null>	<null>	<null>	<null>
7	2003/01/07 00:00:00.000	2003	1	7	<null>	<null>	<null>	<null>
8	2003/01/08 00:00:00.000	2003	1	8	<null>	<null>	<null>	<null>
9	2003/01/09 00:00:00.000	2003	1	9	<null>	<null>	<null>	<null>
10	2003/01/10 00:00:00.000	2003	1	10	<null>	<null>	<null>	<null>
11	2003/01/11 00:00:00.000	2003	1	11	<null>	<null>	<null>	<null>
12	2003/01/12 00:00:00.000	2003	1	12	<null>	<null>	<null>	<null>
13	2003/01/13 00:00:00.000	2003	1	13	<null>	<null>	<null>	<null>
14	2003/01/14 00:00:00.000	2003	1	14	<null>	<null>	<null>	<null>
15	2003/01/15 00:00:00.000	2003	1	15	<null>	<null>	<null>	<null>
16	2003/01/16 00:00:00.000	2003	1	16	<null>	<null>	<null>	<null>
17	2003/01/17 00:00:00.000	2003	1	17	<null>	<null>	<null>	<null>
18	2003/01/18 00:00:00.000	2003	1	18	<null>	<null>	<null>	<null>
19	2003/01/19 00:00:00.000	2003	1	19	<null>	<null>	<null>	<null>
20	2003/01/20 00:00:00.000	2003	1	20	<null>	<null>	<null>	<null>
21	2003/01/21 00:00:00.000	2003	1	21	<null>	<null>	<null>	<null>

6. Filter rows

The screenshot shows the 'Spoon - Transformation 1' window. The 'Execution Results' tab is active, displaying a table with 21 rows of data. The table has columns: #, streamDate, streamYear, streamMonth, streamDay, date, year, month, and day. The data represents a sequence of dates from 2003/01/01 to 2003/01/21. The 'Filter rows' step is highlighted in the process flow.

#	streamDate	streamYear	streamMonth	streamDay	date	year	month	day
1	2003/01/01 00:00:00.000	2003	1	1	<null>	<null>	<null>	<null>
2	2003/01/02 00:00:00.000	2003	1	2	<null>	<null>	<null>	<null>
3	2003/01/03 00:00:00.000	2003	1	3	<null>	<null>	<null>	<null>
4	2003/01/04 00:00:00.000	2003	1	4	<null>	<null>	<null>	<null>
5	2003/01/05 00:00:00.000	2003	1	5	<null>	<null>	<null>	<null>
6	2003/01/06 00:00:00.000	2003	1	6	<null>	<null>	<null>	<null>
7	2003/01/07 00:00:00.000	2003	1	7	<null>	<null>	<null>	<null>
8	2003/01/08 00:00:00.000	2003	1	8	<null>	<null>	<null>	<null>
9	2003/01/09 00:00:00.000	2003	1	9	<null>	<null>	<null>	<null>
10	2003/01/10 00:00:00.000	2003	1	10	<null>	<null>	<null>	<null>
11	2003/01/11 00:00:00.000	2003	1	11	<null>	<null>	<null>	<null>
12	2003/01/12 00:00:00.000	2003	1	12	<null>	<null>	<null>	<null>
13	2003/01/13 00:00:00.000	2003	1	13	<null>	<null>	<null>	<null>
14	2003/01/14 00:00:00.000	2003	1	14	<null>	<null>	<null>	<null>
15	2003/01/15 00:00:00.000	2003	1	15	<null>	<null>	<null>	<null>
16	2003/01/16 00:00:00.000	2003	1	16	<null>	<null>	<null>	<null>
17	2003/01/17 00:00:00.000	2003	1	17	<null>	<null>	<null>	<null>
18	2003/01/18 00:00:00.000	2003	1	18	<null>	<null>	<null>	<null>
19	2003/01/19 00:00:00.000	2003	1	19	<null>	<null>	<null>	<null>
20	2003/01/20 00:00:00.000	2003	1	20	<null>	<null>	<null>	<null>
21	2003/01/21 00:00:00.000	2003	1	21	<null>	<null>	<null>	<null>

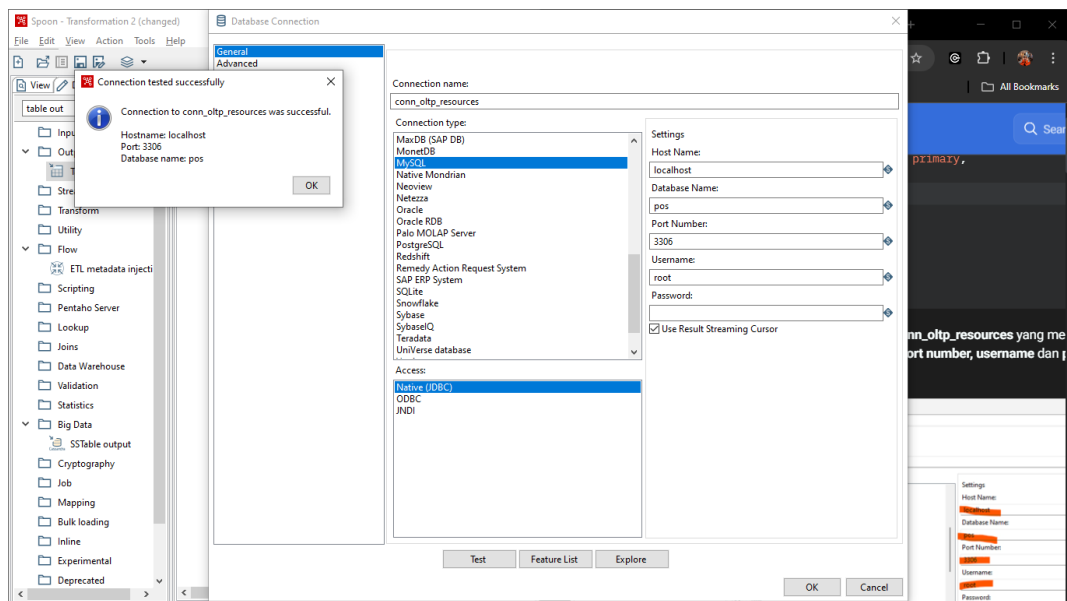
7. Table output

The screenshot shows the 'Table output' step in the SAP Data Services Transformation 2. The 'Execution Results' pane displays a table with 21 rows of data. The columns are: #, streamDate, streamYear, streamMonth, streamDay, date, year, month, and day. The data represents dates from 2003/01/01 to 2003/01/21.

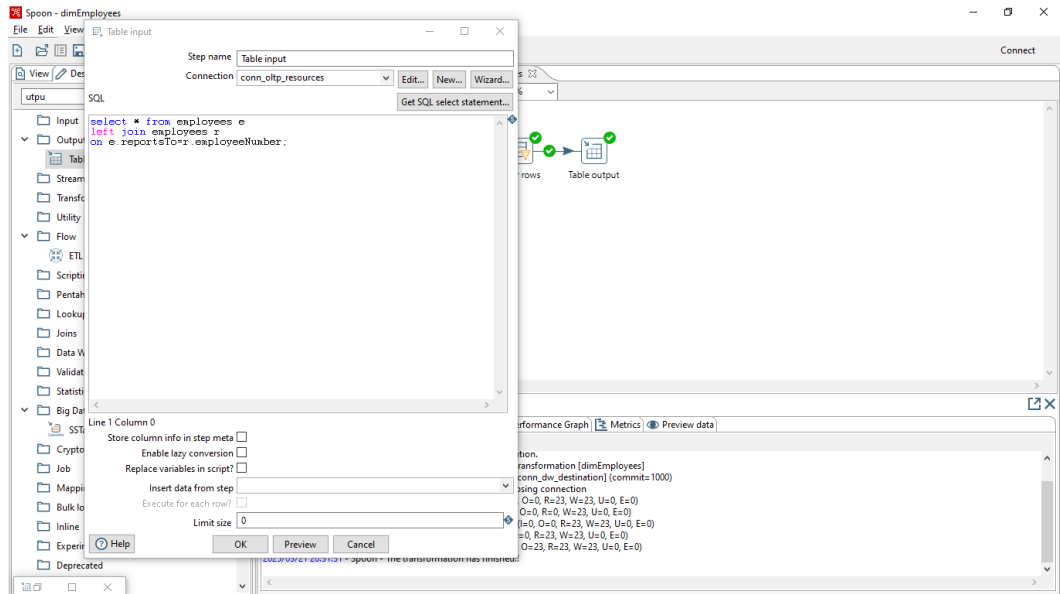
#	streamDate	streamYear	streamMonth	streamDay	date	year	month	day
1	2003/01/01 00:00:00.000	2003	1	1	<null>	<null>	<null>	<null>
2	2003/01/02 00:00:00.000	2003	1	2	<null>	<null>	<null>	<null>
3	2003/01/03 00:00:00.000	2003	1	3	<null>	<null>	<null>	<null>
4	2003/01/04 00:00:00.000	2003	1	4	<null>	<null>	<null>	<null>
5	2003/01/05 00:00:00.000	2003	1	5	<null>	<null>	<null>	<null>
6	2003/01/06 00:00:00.000	2003	1	6	<null>	<null>	<null>	<null>
7	2003/01/07 00:00:00.000	2003	1	7	<null>	<null>	<null>	<null>
8	2003/01/08 00:00:00.000	2003	1	8	<null>	<null>	<null>	<null>
9	2003/01/09 00:00:00.000	2003	1	9	<null>	<null>	<null>	<null>
10	2003/01/10 00:00:00.000	2003	1	10	<null>	<null>	<null>	<null>
11	2003/01/11 00:00:00.000	2003	1	11	<null>	<null>	<null>	<null>
12	2003/01/12 00:00:00.000	2003	1	12	<null>	<null>	<null>	<null>
13	2003/01/13 00:00:00.000	2003	1	13	<null>	<null>	<null>	<null>
14	2003/01/14 00:00:00.000	2003	1	14	<null>	<null>	<null>	<null>
15	2003/01/15 00:00:00.000	2003	1	15	<null>	<null>	<null>	<null>
16	2003/01/16 00:00:00.000	2003	1	16	<null>	<null>	<null>	<null>
17	2003/01/17 00:00:00.000	2003	1	17	<null>	<null>	<null>	<null>
18	2003/01/18 00:00:00.000	2003	1	18	<null>	<null>	<null>	<null>
19	2003/01/19 00:00:00.000	2003	1	19	<null>	<null>	<null>	<null>
20	2003/01/20 00:00:00.000	2003	1	20	<null>	<null>	<null>	<null>
21	2003/01/21 00:00:00.000	2003	1	21	<null>	<null>	<null>	<null>

B. Dimensi Pegawai

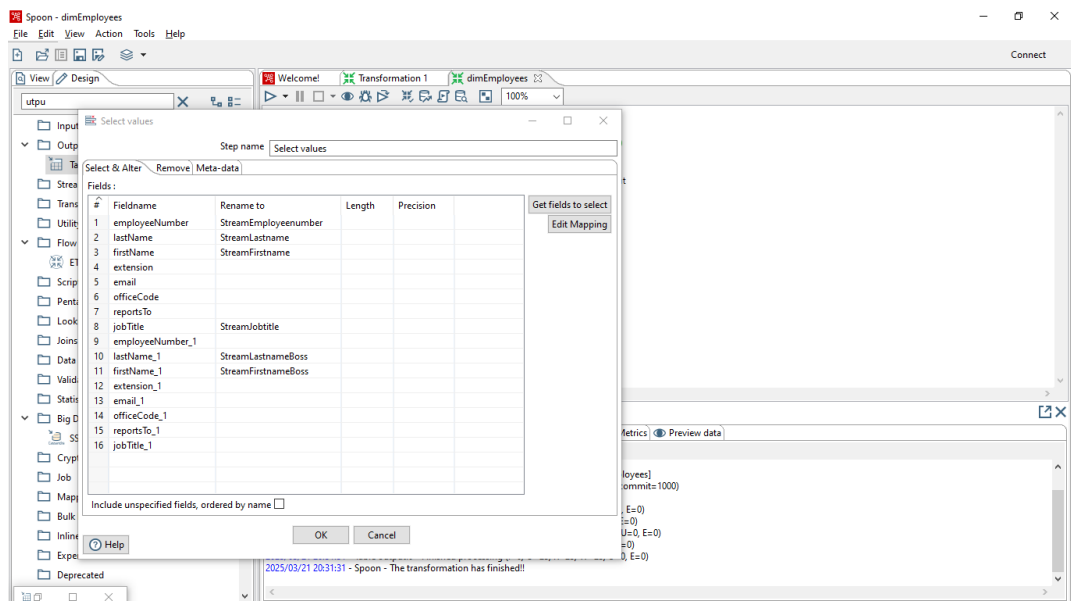
1. Melakukan konfigurasi database connection pada table input

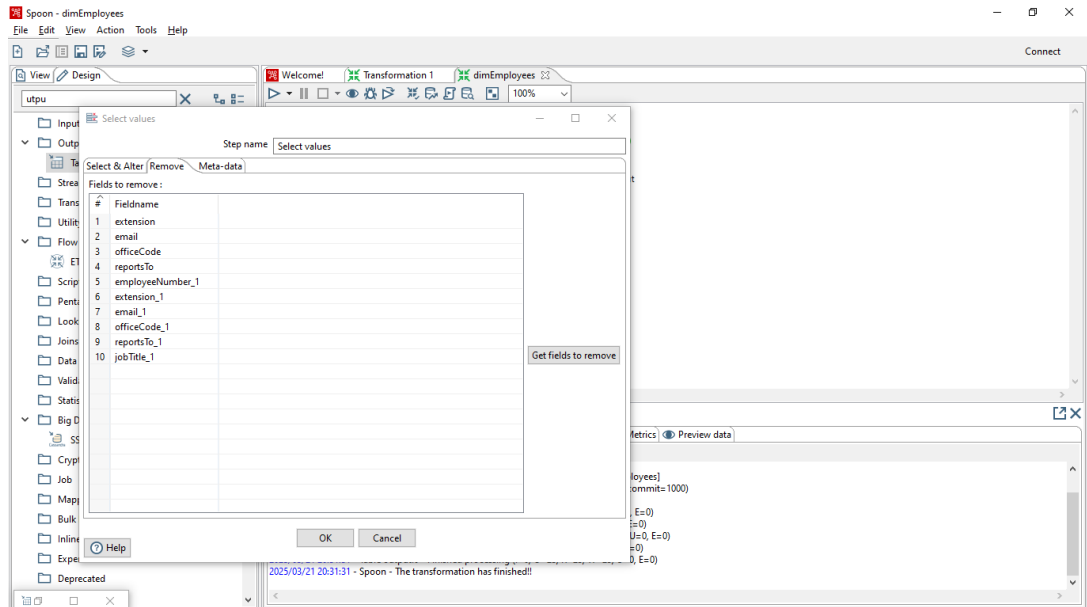


2. Konfigurasi pada table input dengan menghubungkan Connection pada koneksi conn_oltp_resources. Untuk mengambil data sumber

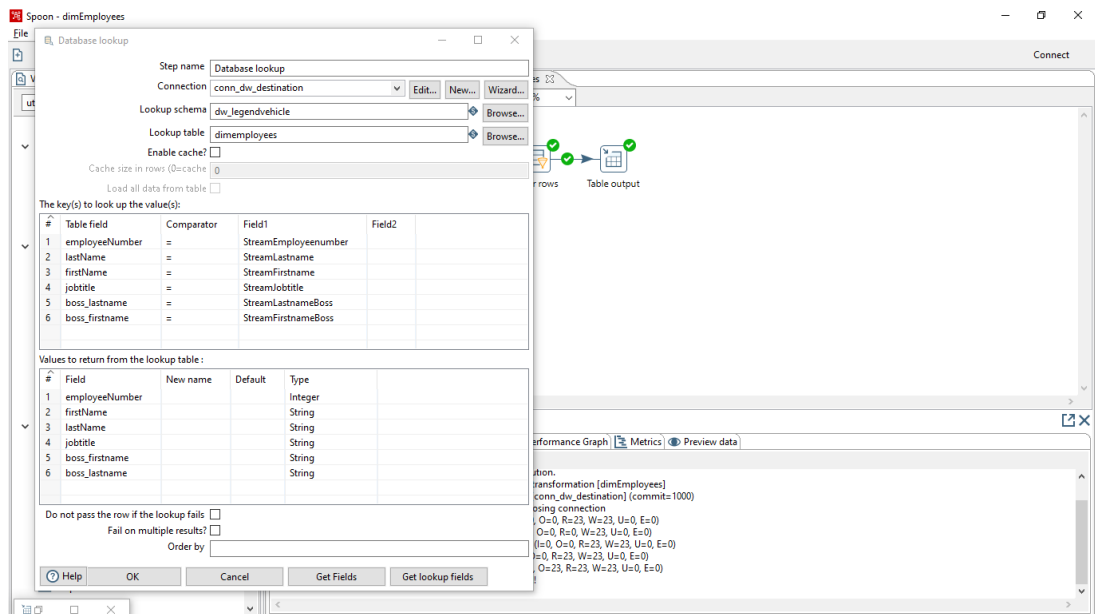


3. Konfigurasi pada Select values yaitu mengambil data dari field employeeNumber, lastname, firstname, jobtitle, lastname_1 dan firstname_1 sebagai data stream yang digunakan pada proses ETL pada tab select & alter.

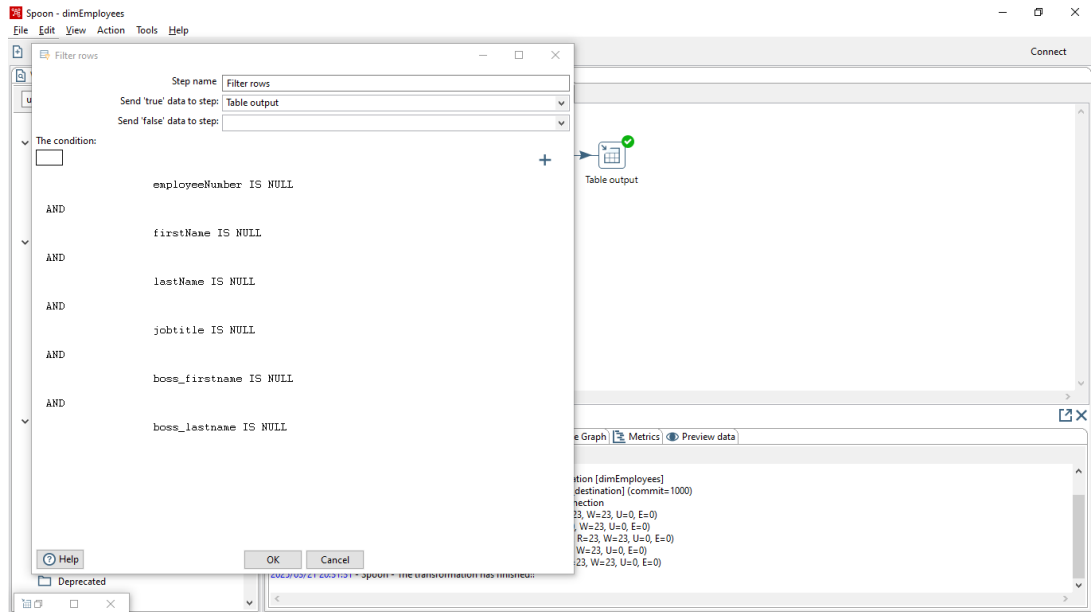




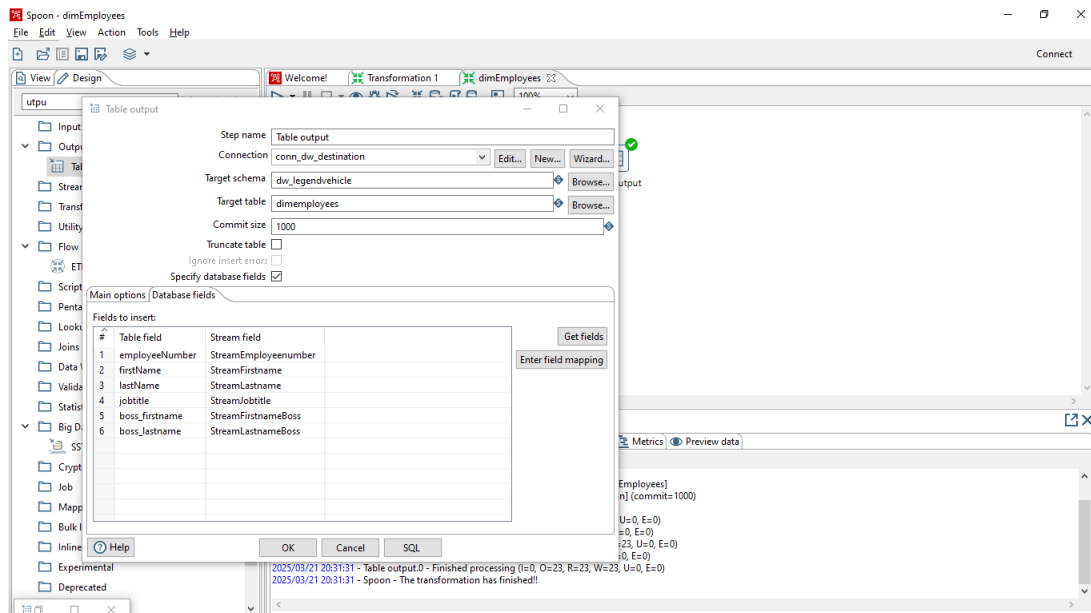
4. Konfigurasi pada database lookup adalah dengan menghubungkan koneksi pada conn_dw_destination dengan table lookup dimEmployees yang telah dibuat pada tahap pertama.



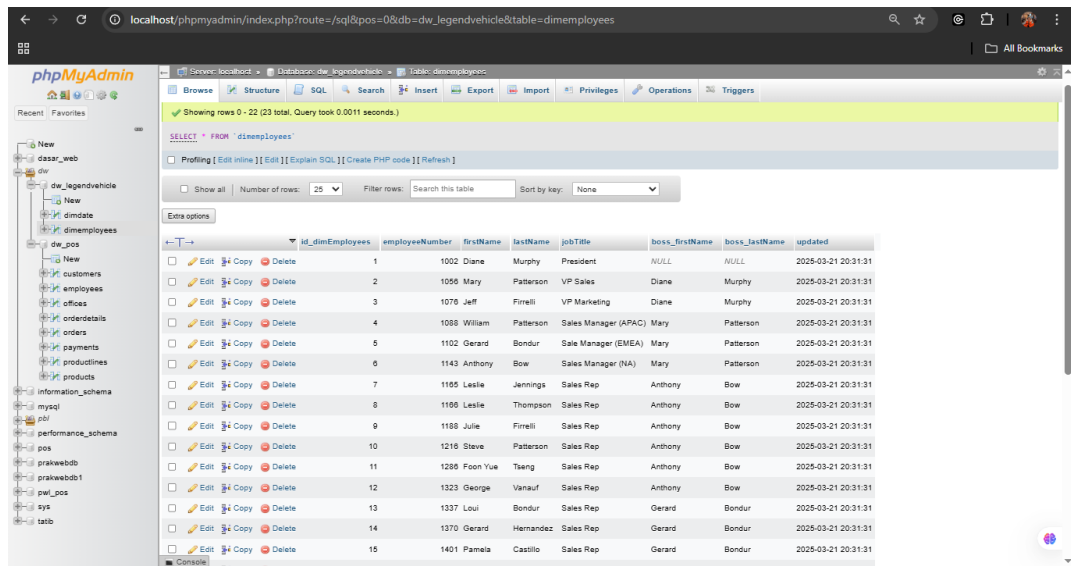
5. Pada filter rows berikan kondisi field yang null pada field dimemployees untuk dimasukkan pada proses selanjutnya. Hal itu menandakan bahwa data stream belum memiliki kesamaan pada data di dimemployees.



6. Table Output



7. tabel dimemployees akan terisi data pegawai dari database OLTP.



Tugas 2:

1. Buka preview tab pada execution result area di setiap proses object. amati input dan output data yang ada. bandingkan di setiap prosesnya. jelaskan perbedaan di setiap prosesnya.

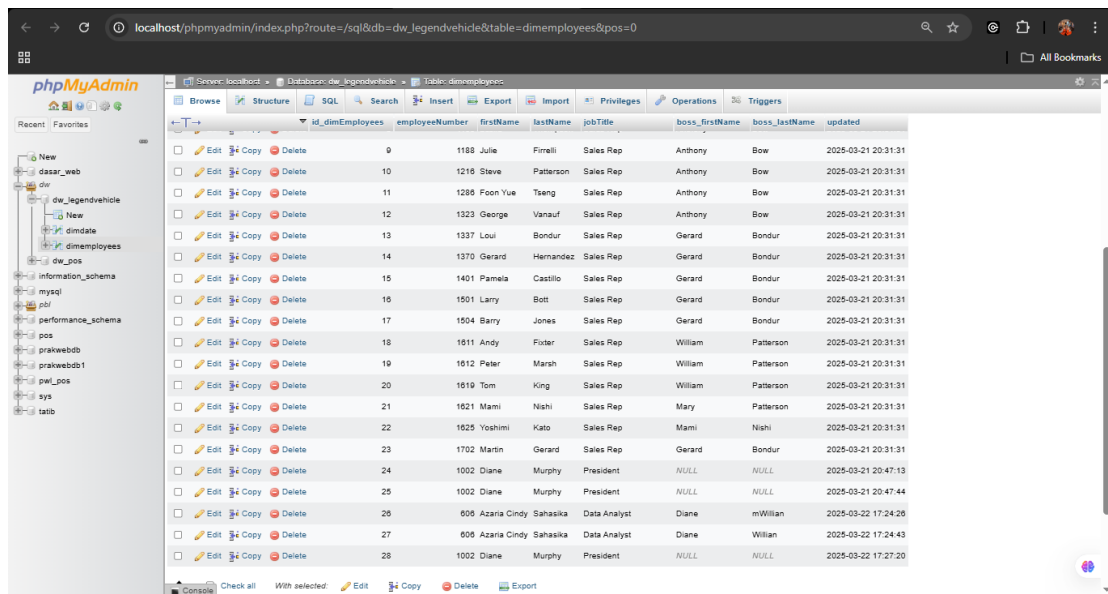
Object Process	Input	Output
Table input	Raw data dari table OLTP	Data stream awal
Select values	Data stream dari table input	Data yang telah difilter berdasarkan field yang dipilih sesuai kebutuhan
Database lookup	Data dari proses select values	Data yang telah dicocokkan dengan table lookup 'dimEmployees'
Filter rows	Data hasil lookup	Data yang hanya berisi nilai null pada 'dimEmployees'
Table output	Data dari proses filter	Data baru ditambahkan ke 'dimEmployees'

2. Jika proses itu di ulangi (di run kembali) apakah data akan redudant?

⇒ Ya, jika proses diulang tanpa validasi, maka data akan menjadi redundan karena proses ETL akan menambahkan data yang sama ke dalam tabel setiap kali proses dijalankan.

3. Tambahkan nama anda pada table employee di OLTP. jalankan kembali transformasi ini. Amati hasilnya, apa yang terjadi?

⇒ Jika data baru dengan nama yang dimasukkan belum ada di dimEmployees, maka data tersebut akan ditambahkan ke dalam table dimEmployees. Jika data sudah ada, maka akan terjadi duplikasi jika tidak ada validasi untuk mencegah pengulangan data.



The screenshot shows the phpMyAdmin interface for a database named 'dw_legendvehicle'. The table 'dimEmployees' is selected, and its structure and data are displayed. The table has columns: id_dimEmployees, employeeNumber, firstName, lastName, jobTitle, boss_firstName, boss_lastName, and updated. The data is as follows:

id_dimEmployees	employeeNumber	firstName	lastName	jobTitle	boss_firstName	boss_lastName	updated
9	1188	Julie	Firrelli	Sales Rep	Anthony	Bow	2025-03-21 20:31:31
10	1218	Steve	Patterson	Sales Rep	Anthony	Bow	2025-03-21 20:31:31
11	1288	Foon Yue	Tseng	Sales Rep	Anthony	Bow	2025-03-21 20:31:31
12	1323	George	Vanauf	Sales Rep	Anthony	Bow	2025-03-21 20:31:31
13	1337	Loui	Bondur	Sales Rep	Gerard	Bondur	2025-03-21 20:31:31
14	1370	Gerard	Hernandez	Sales Rep	Gerard	Bondur	2025-03-21 20:31:31
15	1401	Pamela	Castillo	Sales Rep	Gerard	Bondur	2025-03-21 20:31:31
16	1501	Larry	Bott	Sales Rep	Gerard	Bondur	2025-03-21 20:31:31
17	1504	Barry	Jones	Sales Rep	Gerard	Bondur	2025-03-21 20:31:31
18	1611	Andy	Flister	Sales Rep	William	Patterson	2025-03-21 20:31:31
19	1612	Peter	Marsh	Sales Rep	William	Patterson	2025-03-21 20:31:31
20	1619	Tom	King	Sales Rep	William	Patterson	2025-03-21 20:31:31
21	1621	Mami	Nishi	Sales Rep	Mary	Patterson	2025-03-21 20:31:31
22	1625	Yoshimi	Kato	Sales Rep	Mami	Nishi	2025-03-21 20:31:31
23	1702	Martin	Gerard	Sales Rep	Gerard	Bondur	2025-03-21 20:31:31
24	1002	Diane	Murphy	President	NULL	NULL	2025-03-21 20:47:13
25	1002	Diane	Murphy	President	NULL	NULL	2025-03-21 20:47:44
26	606	Azaria Cindy	Sahasika	Data Analyst	Diane	mWilliam	2025-03-22 17:24:26
27	606	Azaria Cindy	Sahasika	Data Analyst	Diane	William	2025-03-22 17:24:43
28	1002	Diane	Murphy	President	NULL	NULL	2025-03-22 17:27:20